Macro Sizes QUANTGEN's Risk, Never Its Direction

Macroeconomic knowledge belongs in QUANTGEN as a risk-sizing, data-hygiene, hedge-validity and event-handling layer. It should never be a source of direction. Two simulated expert panels, one of professors and one of senior finance executives, reached that answer independently after testing every claim against 1926–2026 data. Both refuted every flagship macro rule in QUANTGEN's prior design. Of 60 linked propositions, **27 became shared integration links, 11 became prohibitions, and only two macro legs survived**, both only as shadow (forward-simulated) cards: USD/KRW up 3% over 20 days scales Korea down to 0.75×, and the US 10-year up 40 bp over 20 days halves US duration-sensitive exposure. What works is sizing a strategy by its own risk state. Own-volatility scaling plus a floored Daniel–Moskowitz bear gate lifts US momentum's Sharpe from **0.40 to 0.80** and cuts its maximum drawdown from −63% to −21%. A Korea-local 20-day volatility target cut the 22 Jun–30 Jul 2026 KOSPI crash from **−38.6% to −12.2%**, a crash that no US-sourced indicator saw. Timing errors, not economics, produced the largest fake result: the old Korea gate's **+0.31 Sharpe fell to +0.05** once US-session data were lagged honestly. Current conditions make the point concrete. 2026 is a real-yield, oil-shock and Korean-leverage regime: the Fed hiked on 16 Sep, the 10-year closed at **5.29%** (highest since May 2002), and KOSPI doubled and then crashed while the won rallied. For the real QUANTGEN, a read-only research desk with a nine-step gate, the main consequence is a mismatch. A de-risk overlay is worth 0.00 to +0.10 Sharpe and would need 33–44 years of data to detect. The gate's benchmark test will therefore reject every overlay both panels endorse. Overlays need their own pre-registered, drawdown-first claim type. Spec v1.1 now defines one, and re-ran the key numbers at QUANTGEN's own costs: momentum own-vol scaling reaches an indicative PASS, the Korea vol target still protects but is INSUFFICIENT DATA, and both macro legs are indicative REJECTs (see the Addendum after the Conclusion).

How to read this report. The evidence base is 13 round-1 position papers, two data labs, a verified 2024–2026 timeline, seven round-2 cross-examinations, two panel-chair resolutions and a round-4 "neural link" synthesis. All panel personas are simulated, fictional archetypes. Their arguments are reported as positions in a structured debate, and no statement here is attributed to a real firm or person beyond its published research. Numbers marked "local" were computed on FRED, Yahoo, Ken French and KOSPI data downloaded on 2 Oct 2026. Their scripts and outputs are in `/home/claude/quantgen_macro_data/` (results files are named in the text). The companion implementation documents are `/home/claude/deliverables/QUANTGEN_MACRO_INTEGRATION.md` (the spec) and `/home/claude/deliverables/QUANTGEN_MACRO_BACKLOG.md` (the backlog). This report summarises them and does not reproduce them.

2026 turned into a real-yield, oil-shock and Korean-leverage regime

The macro backdrop on 2 Oct 2026 is a sequence of three regimes. A model calibrated on any one of them would have mis-specified the next. From January 2024 to mid-2025, disinflation and Fed easing were interrupted by short policy shocks: the 5 Aug 2024 yen-carry unwind and the April 2025 tariffs. From mid-2025 to June 2026, an AI and memory-chip melt-up lifted KOSPI **+197%**, from 30 Jun 2025 to its 22 Jun 2026 closing peak, and a war-driven energy shock arrived on top of it. Since June 2026 the world has been re-tightening: the ECB, BoJ, BoK and Fed all hiked, Korea's leverage unwound, and long-end term premia jumped (local: `T1_events.csv`).

US labour cooled for two years, then re-tightened without a recession

Unemployment rose from **3.7% (Jan 2024) to a 4.5% peak in November 2025** (first print 4.6%, with payrolls +64k), then fell back to **4.1% in July–August 2026** (BLS; BLS). In the current vintage, payroll growth averaged 121.6k a month in 2024, 9.7k in 2025 (the weakest non-recession year in the sample) and 80.4k in January–August 2026. Six months between January 2025 and February 2026 were negative. The vacancies-to-unemployed ratio fell from 1.36 to 0.87 in December 2025 before recovering to 1.01 (local: `A1_labor_dashboard.csv`). The July 2024 print, at 4.3% with +114k jobs (JPMorgan), triggered the real-time **Sahm rule at 0.53**, rising to 0.57 in August. No recession has followed in the 26 months since, and the NBER still lists April 2020 as the last trough (FRED; NBER).

Three data problems now define this labour picture, and each is a trap for any backtest. Payroll benchmarks were revised by **−818k for 2024 and −911k for 2025**, the latter the largest since 2009 (BLS; BLS). The real-time Sahm reading of 0.53 is 0.50 in the current vintage. And the 43-day shutdown meant **October 2025 CPI and household-survey data were never collected** (Richmond Fed). A positional `shift(12)` across that gap silently reports August 2026 CPI inflation as 3.71% instead of the correct 3.35% (local: T1-C5 check on `CPIAUCSL`).

Inflation re-accelerated on energy, and the Fed reversed course

The Fed cut **175 bp between September 2024 and December 2025**, to 3.50–3.75%. It held through July 2026 while dissents migrated from doves to hawks: the April vote was 8–4 and the July vote 9–3, with three members voting to hike (Fed; Fed). On **16 Sep 2026 it hiked 25 bp to 3.75–4.00%, by 12–0**, saying "Inflation remains elevated" and citing geopolitical uncertainty (Fed). The median dot implies **4.1% at end-2026 and end-2027** (Fed press conference). Kevin Warsh was confirmed as chair 54–45 on 13 May and sworn in on 22 May (NPR). Four months into his term the Fed hiked, falsifying the market prior that "the President's pick means cuts".

Headline CPI went from **2.4% in February 2026 to 4.2% in May and 3.4% in August**, with core at only 2.4% and gasoline up 27.4% year on year (BLS). Core PCE is 3.01%. Its three-month annualised rate was already 4.27% in February, before the war, so the core pickup was not purely oil (local: `A3_phillips_monthly.csv`). Seven Taylor-rule variants put the appropriate rate at 4.30–6.21%. Policy therefore sits **0.4–2.3 pp below the rules (central estimate −0.9 pp)**: the hike is rule-consistent in direction, and its size is not identified (Fed Board policy rules; local: `A5_taylor_rules.csv`).

The BoK, BoJ and ECB also tightened into the energy shock, against the textbook advice to look through supply shocks, and the BoE came close. The **BoK** cut 100 bp to 2.50% (October 2024 to May 2025), then hiked to 2.75% on 16 Jul 2026 (unanimous) and **3.00% on 27 Aug 2026 by a 6–1 vote** (UPI; BoK). The **BoJ** went from negative rates to **1.25% on 18 Sep 2026**, a 31-year high (Al Jazeera). The **ECB** hiked from 2.00% to 2.25% on 11 Jun and **2.50% on 10 Sep** (ECB). The BoE held at 3.75% with three votes to hike (BoE).

The long end broke out on real yields and term premium, not inflation expectations

The 10-year Treasury closed at **5.29% on 30 Sep 2026, the highest close since 14 May 2002** and above the 2007 peak of 5.26%. The "highest in about 19 years" description that circulated applies to the 5.0–5.26% levels, not to 5.29% (local: `DGS10`). The 2-year is at 4.88%, the 30-year at 5.64%, and the 10-year TIPS yield at 2.93%, its highest since November 2008. The Kim–Wright term premium is 1.02%, its first reading at or above 1.0 since 2010. Year to date, the 10-year rose 111 bp, of which **TIPS +100 bp and breakevens only +11 bp**. The Kim–Wright split is +45 bp term premium and +54 bp expected path (local: `A7_rates_windows.csv`).

This is the second time in two years that the expectations hypothesis failed for long yields. After the September 2024 first cut, the 10-year *rose* **113 bp while the Fed cut 100 bp**, with the term premium up 76 bp. That was the largest post-first-cut rise in seven easing cycles; the other six ranged from −67 to +42 bp. On 19 Aug 2026 the Treasury announced it would double long-end liquidity buybacks from 9 September (Treasury). Stocks and bonds now fall together. The 63-day stock–bond correlation is **+0.53**, and over 2021–26 bonds rallied by **0.0 bp** on the worst 5% of equity days, against +6.4 bp in 2000–20 (local: `A8_corr_by_year.csv`).

The Hormuz oil shock was the largest supply disruption on record

US and Israeli strikes on Iran on 28 Feb 2026 cut Strait of Hormuz flows to **under 10% of roughly 20 mb/d**. The IEA's record 400 mb coordinated stock release on 11 March did not cap prices (IEA); Dated Brent rose $11.4 the next day. Dated Brent spot went from **$61.35 at end-2025 to $126.69 at end-March and a peak of $138.21 on 7 Apr**. It fell to $70.46 by 30 June after a ceasefire, a US blockade and a June memorandum of understanding. It then rebounded to $130.8 on 15 Sep and **$113.96 on 29 Sep**, as the memorandum's vague strait language collapsed into a tanker war (Columbia CGEP; Al Jazeera).

The often-quoted $127 and $114 are **spot prices**. Front-month futures peaked at $118.35 on 31 March and stood near $103 on 1 Oct, and the spot–futures gap reached about **$29 in April**. Yahoo's continuous Brent future also prints a −14.5% "move" on 1 Apr 2026 that is mostly a contract roll (local: `r3_chair_P_checks.py`). Twenty days into the shock, the 2026 damage was far larger than in 2022: KOSPI −15.5% vs +0.4%, USD/KRW +5.2% vs +2.1%, and the 10-year real yield +32 bp vs −4 bp. The 5-year breakeven rose only 17 bp, and gold *fell* 13.8% (local: `A10_oil_shock_event_paths.csv`).

KOSPI doubled on memory chips, lost 38.6% in 27 sessions, and the won rallied anyway

Korea moved from political crisis to boom to crash in under two years. KOSPI fell 9.6% in 2024, a year that ended with the 3 Dec martial-law shock. It rose **+75.6% in 2025** on governance reform and the memory super-cycle, and **+101.1% in H1 2026**, closing at a peak of **9,114.55 on 22 Jun 2026**. That happened while foreigners were net sellers of **₩148.3tn** and retail investors bought single-stock leveraged ETFs, listed from May 2026 (KEIA). Samsung and SK hynix reached **55.5% of KOSPI market cap**, against 22% at end-2023, and supplied about two-thirds of the 2025–26 index-point move (local: `B3_samsung_hynix_attribution.csv`; Sedaily).

The unwind was structure-amplified. KOSPI fell a record **12.06% on 4 Mar** (Korea Herald) and **9.99% on 23 Jun**, a day on which retail investors bought a record ₩8.54tn (Sedaily). July brought four circuit breakers, 15 sidecars and the margin call of a leveraged AI hedge fund, the last a single-source T2 report (Yahoo Finance). The index troughed at **5,593.56 on 30 Jul**, then posted a record **+17.91% day on 31 Jul** on hyperscaler capex reaffirmations (Sedaily).

From 22 Jun to 30 Jul, KOSPI fell 38.6% while the S&P 500 fell 0.5%, TAIEX 16.4% and the Nikkei 14.5% (15.1% to its 29 Jul low). The same AI shock thus produced about 2.4× the drawdown in Korea. Q3 closed **−19.3%** (not the −18.8% sometimes quoted), and KOSPI stood at 6,943.66 on 2 Oct.

The won broke the textbook link. USD/KRW went from 1,477.9 at end-2024 to 1,555.96 on 5 Jun 2026, then **fell to 1,356.5 by 25 Sep**: −12.8% from that high and −12.4% over Q3, while the broad dollar fell only 0.5% and foreigners sold ₩21.5tn of Korean equities in September. Record exports, **$120.9bn in September (+83.5%)** with chips above $60bn, together with BoK hikes and the NPS swap line, made the won behave like an exporter-earnings currency rather than a risk proxy (Korea Times). The yen is near 157 after two solo interventions in 2026 and a joint US–Japan intervention on 30–31 Jul, the first since 2011 (MoF). Bitcoin fell 53% from its October 2025 peak to $58.6k on 30 Jun 2026, including −33% in H1 2026 while the war was under way, and has rebounded to about $84.8k (CoinDesk; local: `CBBTCUSD`).

State on 30 Sep–2 Oct 2026	Reading	What it means for the overlay
Fed / BoK / BoJ / ECB	3.75–4.00% / 3.00% / 1.25% / 2.50%	Synchronised tightening; policy-direction states must allow sign flips
UST 10y; 20-day change	5.29%; +50 bp	Shadow rates leg ON in data dated 23 Sep; under strict H.15 timing it applies from the 25 Sep close
10y TIPS / breakeven / KW term premium	2.93% / 2.36% / 1.02%	Real-yield and term-premium shock, not de-anchoring
Core PCE; stock–bond corr (63d)	3.01%; +0.53	Hedge credit for bonds OFF
VIX / VIX3M; HY OAS 21-day change	16.34 / 18.37; +47 bp	No entry block; credit just under the +50 bp context threshold
USD/KRW 20-day change	−1.8%	Shadow KRW leg OFF
KOSPI; 21-day realised vol; Korea vol-target weight	6,943.66 (−23.8% from peak); 31.9%; 0.61	Local risk control still binding
Samsung + SK hynix share of KOSPI	≈53.7% implied	Pair-concentration cap active (>45%)
Dated Brent / front-month futures	$113.96 / $102.59	Name the series in every rule

Sources: local `A12_macro_snapshot.csv`, `R3_CHAIR_E_checks.txt`*; spec §11.*

Textbooks teach the mechanisms; finance curricula teach why they fail

The professors mapped the attached textbooks and about 20 university courses onto the macro-to-markets channels. They found a clean division of labour.

Textbook macro supplies the mechanisms. Lee & Kim's *Dynamic Macroeconomics* (Korea University slides; local) covers Korean cycle facts (ch.1), growth (ch.2–5), RBC (ch.6), money (ch.7), the New Keynesian Phillips curve (ch.8), and NK-DSGE with Taylor rules and commitment versus discretion (ch.9). Blanchard & Johnson 6e (local) adds Okun's law, the expectations-augmented Phillips curve with its 1.8 sacrifice ratio, the term structure and stock pricing (ch.15), and the open-economy models of UIP, Mundell–Fleming and sudden stops. Kirchgässner–Wolters–Hassler (local) supplies the econometric toolkit: VAR, Granger causality, unit roots, cointegration and GARCH. Dutta (local) gives the game theory of commitment, OPEC-style cartels, tariffs and wars of attrition.

Finance curricula supply the failure analysis, and none of the four textbooks contains it. Cochrane's free Booth/Hoover class has modules on time-series predictability and on "Equity Premium, Macroeconomics, Asset Pricing" (Cochrane).

Princeton's ECO 529 centres risk-bearing capacity and intermediary asset pricing (Princeton). The Stanford/Columbia international-finance syllabus teaches Meese–Rogoff, carry and CIP deviations as standing puzzles (Maggiori–Schreger). Berkeley's monetary lecture treats the 2021–23 deviation from the Taylor rule as the Fed falling "behind the curve" (Steinsson L17). Oxford's MFE notes provide the most directly usable layer for QUANTGEN: bootstrap, GARCH, realised variance and VaR/ES backtests (Sheppard).

One correction matters for self-study. MIT 14.02 now assigns **Blanchard 8e**, not the 6e on GEON's shelf, so courses should be mapped by chapter title (MIT OCW). Korea University's published course descriptions cover macro dynamics (ECON 359), money and banking (ECON 331), international financial economics (ECON 461) and time-series econometrics (ECON 242). They list **no empirical asset-pricing, predictability or systematic-trading course** (KU); KAIST's MFE does, in FE537 Statistical Arbitrage and FE547 Algorithmic Trading (KAIST). The macro-to-trading bridge must therefore come from outside the KU core. The panel's recommended order of study is: volatility econometrics, then out-of-sample predictability hygiene, then strategy families, then monetary-event reading, then FX, and macro-finance theory last.

Knowledge block	What it teaches	Linked real data (local test)	What the data said	Where it enters QUANTGEN
L&K ch.8–9; B&J ch.8; Hazell et al. (NBER)	NKPC, Taylor principle, commitment	PCEPILFE on unemployment gap, by era; 7 Taylor variants	Slope −0.74 → −0.18 → +0.04; policy 0.4–2.3 pp below rules	Context log only; NK as scenario sign templates
B&J ch.2; Ball–Leigh–Loungani (NBER)	Okun's law	GDPC1 vs UNRATE, rolling	β −0.407 → −0.05; break ≈2011	Prohibited as a coefficient
B&J ch.15; Estrella–Trubin (NY Fed)	Expectations hypothesis, curve as a recession predictor	T10Y3M, probit, Kim–Wright term premium	74% probit with no recession; 10y +113 bp after the first cut	Real yield and term premium logged; curve as prior
B&J ch.18–21; Maggiori–Schreger syllabus	UIP, sudden stops, carry	Fama regressions, carry drawdowns	USD/JPY b −1.92 → 0.11; KRW R^2 0.022	No rate-differential FX model; carry sized like an option seller
KWH; MIT 14.384 (OCW); Oxford MFE	Granger, unit roots, breaks, GARCH	Granger tests on IP; monthly ARCH	Returns lead IP (p 2e-8); IP → returns p 0.04; GARCH β 0.75	Volatility module; validation rules
Cochrane M10; Koijen–Van Nieuwerburgh (Columbia)	Predictability is small and break-prone	Term spread → next-month return	Out-of-sample R^2 −0.57% (1980+), −1.28% (2010+)	Out-of-sample R^2 gate before any predictor scales weights
Campbell–Pflueger–Viceira (NBER)	Inflation regime sets the stock–bond sign	Daily corr by core-PCE bucket	−0.29 / −0.11 / +0.21	Hedge-validity state
Dutta; Jarociński–Karadi (AEJ); Stein 2009	Credibility, information shocks, crowding	FOMC classes 2021–26; 5 yen interventions	INFO_NEG days +3.99% over 10 days; intervention vol rose in 4 of 5	Logged; FX-intervention event rule for a future FX sleeve
Pedersen (NYU Stern); D&M; Barroso–Santa-Clara	Strategy families and their crash risk	French daily factors 1990–2026	Own-vol scaling 0.40 → 0.91 (cap 2)	Momentum sizing (BUILD NOW)
ECO 529; Cochrane "Macro-Finance" (NBER)	Risk-bearing capacity varies	No KRX margin or flow data locally	Untested	Korea leverage state (backlog P1)

Read as a whole, the map says the textbooks are the right lens for *reading* central banks: why the Fed hiked into an oil shock, and why commitment keeps the front end high after a cost-push shock. They are the wrong source of *coefficients*.

Every fixed textbook relation the panels re-estimated broke inside QUANTGEN's sample.

Where the models work, and where they broke between 1960 and 2026

The models fail in three recurring ways.

Structural breaks. Relations estimated under one policy regime die under the next, exactly as the Lucas critique in B&J (p.176) warns. Okun broke around 2011. The Phillips curve flattened after 1985 and vanished after 2008, consistent with anchored expectations (Hazell et al.). The 1.8 sacrifice ratio implied about 5.5 pp-years of excess unemployment for the 2023–25 disinflation; the realised figure was −1.57, with unemployment *below* its natural rate (local: `A3_phillips_fits.csv`).

Lateness and revisions. Recession signals arrive after the fact or get revised. The Sahm rule fired a median **two months after** recession starts in 9 of 9 cases since 1960. NBER announced the April 2020 trough about 15 months later. Chauvet–Piger's real-time model calls peaks about a month after NBER's own announcement (Chauvet–Piger).

Regime-conditional signs. Some signs hold only within a regime and flip without warning. The dollar's haven bid failed in April 2025, the stock–bond correlation turned positive with inflation, and the won rallied during a foreign sell-off.

Underneath all three sits a cleaner empirical law that both panels adopted. **Market-priced stress variables forecast variance, not the mean.** The VIX/VIX3M slope predicts 21-day realised volatility with t = 7.17 (R^2 0.385) and next-21-day returns with t = 0.04. Across 22 episodes since 1988 in which Brent rose 30%, forward 63-day returns were 2.67% against 2.66% unconditional. Weeks with the dollar and equities both down were followed by **+7.27%** over 13 weeks, against 2.84% normally (local: `r2_P2_p3_checks.py`, `r2_P2_p1_checks.py`, `r2_P3_p4_checks.py`).

Model	Works when	Failed when (dates, numbers)	Why	QUANTGEN treatment
Okun's law	US 1961–2010, β −0.407 (t −10.8)	2011–19 β −0.05; QLR break ≈2011; ex-Covid rolling R^2 ≈0.03; 2023–25 Δu +0.88 pp on 2.7% GDP growth	Labour hoarding, productivity and labour-supply swings, vintages	Prohibited as a coefficient; context only
Phillips curve / NKPC / sacrifice ratio	1961–84 slope −0.74	1985–2019 −0.18; 2011–25 +0.04; out-of-sample RMSE 2.15 vs 1.79 for no-change; Sep-2022 fit 2.42% vs 5.64% actual	Anchored expectations; supply shocks; v/u non-linearity (Benigno–Eggertsson)	Inflation state = realised core momentum + expectations
Taylor rule	1990–2007 RMSE 1.7–1.8 pp	Rate 5.1–6.6 pp below the rule on average in 2021–22 (up to 9.6 pp in Mar 2022); ELB 2009–15; now 0.4–2.3 pp below	ELB, look-through choices, r* and gap specification (Fed)	Range-reported context; never a threshold
Sahm rule	9 of 9 recessions since 1960	Median +2 months *after* the start; Jul-2024 0.53 then 26 months without recession; real-time 0.53 vs current 0.50	Calibrated on demand recessions; revisions (Sahm)	Logged prior
Yield-curve probit	Inversion preceded every recession 1968–2019	Record 534-day 10y–3m inversion; probit 0.744 in May 2023; no recession through Sep 2026; 10y–2y false in 1982, 1998, 2022–24	Expected cuts; compressed term premium; only 6 onsets in sample (FEDS Note)	Prior only; drop 10y–2y
Expectations hypothesis / term premium	Rough fit in 1990–2007 easing cycles	Sep-2024→Jan-2025: 10y +113 bp with 100 bp of cuts (TP +76 bp); 2026 TIPS +100 vs BE +11	Fiscal supply, inflation uncertainty	Track real-yield and TP changes, not the policy direction
UIP / carry	Long-run tendency; USD/JPY carry SR 0.24 (0.46 in 2015–26)	Fama b −1.92 (1991–2007); KRW monthly R^2 0.022; carry −10.6% in 18 days (10 Jul–5 Aug 2024) while VIX stayed at 12.85–16.36 until 31 Jul; long-KRW −36.3% in Dec 1997	Time-varying risk premia; crowded funding (Fama; BNP; BIS)	No rate-differential FX model; carry gated on FX trend and JPY vol, not VIX
Dollar safe haven	Broad USD up in 10 of 11 S&P drawdowns since 2007	Apr-2025: DXY −3.9%, 10y +28 bp, S&P −5.4%; 2025–26: USD down in 57% of worst-decile weeks (n = 7)	US-originated policy shocks	Classic risk-off needs USD up AND yields down; USD↓ with equities↓ = hedge-failure flag
Gold vs real rates	Change β −11.6%/pp (2003–21), −7.6%/pp (2022–26)	2006–21 level fit implies $626 vs $4,187 actual; gold −13.8% in the 2026 oil shock	Level break unidentified (Chicago Fed; Erb–Harvey)	Short-horizon beta only
Stock-bond correlation	Negative every year 2007–21	2022 60/40 −17% to −23%; Apr-2025 10y +33 bp; Q3-2026 10y +85 bp; 63d corr +0.53 now	Inflation regime and real-rate shocks	Hedge credit off when core PCE ⩾3% or corr63 >0
Return predictability	In sample 1960–89	Term spread OOS R^2 −0.57% (1980+) and −1.28% (2010+); D/P evidence "disappeared after 1995"; over 1/3 of 46 predictors lose significance (Goyal–Welch–Zafirov)	Breaks in means; persistence bias	OOS-R^2 gate before any predictor scales a weight

Model	Works when	Failed when (dates, numbers)	Why	QUANTGEN treatment
Regime models (Markov / HMM / jump)	Smoothed US SR 1.518 (uses future data)	Real time: 16.9 switches a year, 49% false (US); 30.8 a year, 77% false (KOSPI); SR 0.390 vs buy-and-hold 0.539	No persistence penalty; smoothing look-ahead (Shu–Yu–Mulvey, T1-pre)	Filtered states only; no unpenalised daily HMM gates
Macro factor timing	In sample only	Local: curve cut −0.09, real-rate gate −0.04, credit −0.05 (2020–26), four-state table −0.07; timing IR 0.25–0.50 vs a 1.8 bp break-even (Ilmanen et al.)	Unstable low-R^2 relations; costs	Prohibited
Street forecasts	Small average drift	S&P consensus abs error 17.13% (2022–25; Avantis); 2023 recession call Brier 0.49; 2026: cuts forecast vs the hike, a 10y call of 4.35% by Q4 (JPM outlook) vs 5.29%, Brent $56–58 vs a $93 average; a further −10% dollar call (Morgan Stanley) vs DXY +3.9% YTD	Anchoring, herding, unforecastable supply shocks	Not inputs; SEP error bands (±1.4 pp) as Monte Carlo priors
Surveys and positioning	ACWI −2.4% at 1 month after sell signals	Aug-2026 FMS published 18 Aug, after KOSPI −25.4%; AAII contrarian SR 0.29 vs 0.36 buy-and-hold	Publication lag; tiny n	Context with publication timestamps

Local evidence: `A2`–`A11`, `B1`, `B4`, `E7_street_forecast_audit.csv`, `R2P1_*_realtime_ms_prob.csv`.

What survives this record is short and consistent. **Own-volatility scaling of momentum** works because momentum's risk is persistent and forecastable while its mean is not. It lifts the Sharpe ratio in every region and in 86% of years, and it clears a Bonferroni-120 hurdle (z 3.9–4.35) (Barroso–Santa-Clara; Moreira–Muir). The decay warning is real, though. Its local gain (with a 2× leverage cap) fell from +0.66 in 1990–2011 to +0.26 in 2012–26, and Cederburg et al. find that 103 vol-managed strategies "do not systematically outperform" (Cederburg et al.).

The **Daniel–Moskowitz bear-state gate** adds about +0.15 on top in the published paper. Locally, it misses crashes that follow bull markets: Nov-2020 (−12.0%), Jul-2026 (−10.9%) and Jan-2001 (−21.9%) (Daniel–Moskowitz JFE).

Hedge validity is a state, not a constant. Bonds failed to hedge equities in 2022, April 2025 and Q3 2026, and the dollar failed in April 2025.

How the panels argued: statistics versus survival, and both changed their minds

The professor panel had six members: macro theory (P1), time-series econometrics (P2), asset pricing and factors (P3), game theory and policy (P4), curricula (P5) and quant ML methods (P6). The executive panel had seven: cross-asset strategist (E1), NYSE trader (E2), FX/rates trader (E3), Asia/EM portfolio manager focused on Korea (E4), CRO and systematic PM (E5), IB vice-chair (E6) and global head of research (E7). In round 2, each member cross-examined colleagues *within* the same panel. Reviewers had to give every claim a verdict, re-open the sources of at least eight key numbers per paper, and recompute at least three of them from local data. Each chair then issued a resolution. The professors' chair produced 30 consensus statements, 11 contested and 32 refuted or withdrawn. The executives' chair produced 30, 11 and 25, plus a BUILD NOW / SHADOW / BACKLOG / DROP triage.

The fault line ran between statistical power and survival. The professors demanded multiplicity-adjusted evidence: $z \geq 2.7$ with no more than 8 pre-registered cells, or $z \geq 3.35$ against a Bonferroni-120 bar (Harvey–Liu–Zhu). The executives accepted small, plausible de-risk effects and controlled the risk of being wrong through governance. The cross-asset persona's rebuttal to its own reviewer became the executive panel's motto: "**a one-person shop dies of tails, not of 0.09 Sharpe**". The two positions were reconciled by arithmetic everyone accepted. With overlay and base books about 0.98 correlated, the standard error of ΔSR is 0.034 for 36 years of US data and 0.052 for 11 years of KOSPI. Detecting a +0.05 gain would therefore take **33–44 years** (executive chair resolution, EC-E-02). Every one of 26 tested overlays had a 95% confidence interval that straddled zero.

The disputes that mattered, and how each resolved

The Daniel–Moskowitz floor split the panels in an unexpected direction. The asset-pricing professor argued for 0×. The risk officer argued for 0.5, on the ground that an overlay should never impose a hard zero. Q1b's numbers sided with the professor on Sharpe: +0.26 at 0× against +0.14 at 0.5×. The round-4 recomputation then showed that the floor-0.5 gate still adds **+0.084** on top of capped vol scaling, and +0.056 with 2008–09 removed (share 0.97). Its cost is concentrated in May–Nov 2008, when it earned +9.5% against +21.8% for the static book. Both chairs built it at 0.5 and sent 0.25 to the backlog.

Reversal in fast stress produced a role swap. The professors held gross data showing reversal earns 72% a year at 30.6% volatility on VIX-inverted days, so variance rises faster than the mean. On that basis they said reversal should be held at 1.0× and *never cut*. The executives cited KRX price limits, 41 sidecars in H1 2026 and spreads widening about 2.5× in stress, and wanted it blocked. A read-only desk has no fill data, so the question goes to a forward-simulated test priced with KIS quote snapshots (BL-C02).

The macro-conditioned overlay floor pitted P1 against P6. P1 showed that own-market de-risk overlays earn more when core PCE is at least 3% (+0.14 vs +0.05) and proposed a floor of 0.5 in tightening regimes. P6 replied that every confidence interval contained zero and that macro would have mis-called 2026, when KOSPI rose 68% from 31 Mar to 30 Jun during the Fed's hawkish turn. The resolution split the verdict. P6 won the narrow question of daily equity gross. P1 won on scope: shock type determines which hedges work. The interaction is pre-registered for a locked 2027+ holdout.

Korea discrete flags showed how fragile local rules are. P6's proposed fix, "high vol AND below the 200-day average", removed the 2026 bull tax but also all Q3 protection (−38.1% vs −9.6%). P6 conceded: "exactly why I refused to tune it: every fix is a trial, and none beats buy-and-hold."

The FX desk's KRW refinements all failed: a two-speed gate (+0.002), a residual filter (−0.03) and a "dollar regime B" switch that was on for 1% of days and missed April 2025. The plainest rule survived instead.

Oil was argued out on the 2026 path. The Brent 20-day >+25% gate first fired on 6 Mar 2026, **two sessions after the −12.06% day**, and lowered Sharpe. The textbook AS-shock sign held for headline inflation but not for equities over the following quarter.

Numbers the debate corrected, and which must not propagate

The cross-examination's most durable output is a list of corrected anchors. Several came from the shared briefing itself.

Claim as first circulated	Corrected value	How it was settled
Daniel–Moskowitz dynamic Sharpe "≈1.43 vs 0.71 static"	**0.682 → 1.041 (constant-vol) → 1.194 (dynamic, out of sample), 1934–2013, *JFE* Table 7**. 0.71 is the Table 1 static figure for 1927–2013 ("approximately doubles" was read as 1.43). Q1b's 0.283 → 0.646 is the US 1972-start row of Table 11	Chair re-opened the published PDF (JFE 2016); ≈70% of the gain is plain vol scaling
Prior Korea gate (SOX 60d × USD/KRW 20d) worth **+0.31 Sharpe**	**+0.05** implementable (share 0.67); **−0.16 in 2024–26**; SOX leg alone −0.22	The +0.31 used the US session that closes after KOSPI's close (local: `B3_korea_gate_backtest.csv`)
KRW leg +0.07 using H.10 as "daily, T+1"	**+0.03** (0.5× leg) with Yahoo timing; H.10 is published weekly, on Mondays at 4:15 pm. The spec's 0.75× leg gives +0.016 with one extra session and **+0.001 under strict H.10 publication** (spec v1.1, R4b)	H.10 page re-fetched (Fed H.10)
Rates leg +0.041, full sample	+0.041 is the R2 construction; R4 gives +0.038 on value timing and **+0.010** at a one-business-day lag, which is still 15 minutes early for a decision at the close. At the strict two-business-day lag it is **−0.000** (spec v1.1, R4b). 2015–26: +0.066 to +0.071 on value or live-proxy timing, +0.053 strict	Round 4 and R4b (Fed H.15)
KOSPI Q3 −18.8%; peak ≈9,386	**−19.3%; closing peak 9,114.55 (22 Jun)**	Recomputed from closes
Brent $127 / $114 as tradable	Dated **spot**; futures $118.35 / $102.59	Series named
BoK 27 Aug "unanimous"	**6–1**, base rate 3.00%	BoK release
10y 5.29% "19-year high"	Highest close since **14 May 2002**	DGS10
Fed "≈140 bp behind Taylor"	**−0.89 pp** central; 0.4–2.3 pp range (used realised core PCE, not the SEP projection)	Recomputed
Ilmanen "30% out-of-sample decline"	Value −49%, momentum −23% (n.s.), multifactor −18%	JOIM PDF re-fetched
KRW carry skew −4.06, worst quarter −58.6%	**Skew −2.38, worst −36.9%** (sign error on loss); post-float Fama β −1.63	Recomputed
Lee & Kim ch.9 slide 6: "the textbook CB prioritises inflation"	The slide's rule reacts to output's deviation from steady state, not the gap: Galí's "respond to the right gap" result	P1 re-read slide 5–6
9 Mar 2026 KOSPI −8.11%	Close **−5.96%**; −8.11% was the intraday circuit-breaker print	Closes, not prints

Two concessions shaped the final design more than any single test. The macro theorist withdrew the real-rate momentum gate, which cut momentum during its 2022 rally (−7.2% relative), and the "2026 is a bear steepening on inflation" framing, since the real/breakeven split is about 90/10. The cross-asset strategist withdrew the SOX Korea gate, the VIX momentum switch and the cut-and-re-add boundary behaviour (−0.088 Sharpe), keeping only the drawdown ladder.

The neural link: 27 shared links, 11 prohibitions and two shadow legs

Round 4 mapped all 139 chair items onto **60 single-proposition nodes**, joined by 240 edges: 120 supports, 48 refutes, 44 qualifies and 28 conflicts. Each node was then classed by joint panel strength and data support (local: `R4_neural_link_matrix.csv`). The result is **27 CORE-LINKs** routed to the spec and **11 CORE-PROHIBITIONs** routed to the DO-NOT-BUILD list. Eight of the prohibitions were refuted by both panels; three were refuted by one panel plus the data, with the other panel silent. The remaining 22 nodes went to the backlog: **15 contested, 4 one-sided and 3 untested**. Focus intensity counts the distinct claims each panel devoted to a node, so it measures attention, not agreement. It shows where the effort went: validation, data hygiene and events, not macro signals.

Node	Topic	Focus (Prof / Exec)	Class
N-50	Validation and governance	19 / 13 = 32	CORE-LINK
N-56	Forecasts, surveys, positioning as timing signals	8 / 20 = 28	CORE-PROHIBITION
N-60	Corrected anchors and citations propagate	14 / 10 = 24	CORE-LINK
N-04	Stress variables → variance and entry block	13 / 10 = 23	CORE-LINK
N-37	Event windows, no blackouts	9 / 12 = 21	CORE-LINK
N-40	US overnight → KOSPI opening-gap input	11 / 10 = 21	CORE-LINK
N-48	Quant Lab tails including 2026	7 / 14 = 21	CORE-LINK
N-39	Decision-day momentum size-down	9 / 12 = 21	ONE-SIDED (mostly executive dissent)

The **core links** cluster in four areas.

Size by own risk, not by macro forecast. Overlays are de-risk-only: multipliers stay within [0.5, 1.0], the vol-target weight is capped at 1.0, and only the fail-closed risk engine can go to 0×. Momentum gets own-vol scaling and the D&M gate floored at 0.5. Reversal and quality stay at 1.0×. Stress variables feed volatility estimates and new-entry blocks, never cut-and-re-add triggers.

Korea is its own market. Korea gets a local 20-day vol target capped at 1.0. If Samsung and SK hynix together exceed 45% of KOSPI, they are held to no more than 35% of the long book. US overnight moves enter only as opening-gap inputs: the prior US session explains about 15% of KOSPI daily variance, and the SOX beta rose from 0.17 (2015–19) to 0.56 in 2026 (0.25 over the full 2015–26 sample).

Data hygiene is the largest single source of fake edge. Every input carries UTC and publication timestamps. Lags are calendar-based. Brent is always named as spot or futures. Closes are used, never circuit-breaker prints. Missing inputs fail closed.

Macro as context and hedge-validity state. Sahm, the curve probit, Taylor ranges, real yields and term premium are logged and never act. Hedge credit is withdrawn when core PCE is at least 3% or the 63-day stock–bond correlation is positive.

The **eleven prohibitions** cover every flagship of the prior design: upward multipliers and the four-state table; VIX-term-structure momentum switches; macro factor gates (curve, real rate, credit, levels); Korea gates built from US inputs; fixed textbook coefficients; the lagged oil gate; net-liquidity and M2 rules; regime and ML shortcuts (smoothed probabilities, unpenalised HMMs, "complexity" timing, LLM backtests before the model's training cutoff); Street forecasts and surveys as inputs; 24-hour blackouts; and calendar and issuance tilts.

The "complexity" result deserves mention because it is recent and influential. Kelly–Malamud–Zhou's random-feature macro timing reduces almost entirely to vol-timed momentum: the correlation is 0.93, and its alpha has t −0.12 after a momentum control. That is consistent with Nagel's critique (Nagel).

Round 4 also recomputed the "golden values" under the spec's own caps, and the headline numbers shrank. Momentum own-vol scaling with the leverage cap at 1.0 instead of 2.0 gives **0.40 → 0.71**, and adding the D&M gate at a 0.5 floor gives **0.80**, with maximum drawdown −63% → −21%. The three-factor US book goes **1.36 → 1.75**. The Korea vol target is Sharpe-

neutral over 2016–26 (0.638 vs 0.634) but carries a clear cost in 2016–25 (ΔSR −0.077, share 0.06). It must therefore be judged on drawdown: Jul-2026 −12.2% vs −38.6%, Mar-2026 −7.9% vs −19.9%, against a bull tax in H1 2026 of +51.7% vs +101.1%. The KRW leg at 0.75× adds **+0.032** (share 0.965), and +0.042 on top of the vol target, gross and with DEXKOUS used by value date; under strict H.10 publication it adds +0.001, and net of 35 bp it loses 0.008 (see the Addendum). The rates leg is worth +0.010 with DGS10 lagged one business day over the full sample. That is still 15 minutes early for a decision at the close; at the strict two-business-day lag it is −0.000, so it requires a live yield feed (local: `R4_golden_values.csv`, `R4b_golden_values_net.csv`).

What remains one-sided or contested is the research agenda. Each item already has a pre-registered settling test in the backlog.

Node	Item	Who holds it	Settling test
N-22	USD/KRW 20d >+3% → Korea 0.75× (SHADOW)	Executives (moderate); professors split	Spec v1.1: protective checks PC-1..PC-6 at QUANTGEN's cost and at 2×, ⩾ 30 pooled onsets, ⩾ 12 months of forward-simulated record, on true-timestamp KIS/BoK closes (backlog BL-S01)
N-24	US 10y +40 bp/20d → US duration gross 0.5× (SHADOW)	Executives (moderate); professors weak	Spec v1.1: protective checks at 50 and 100 bp per side under strict H.15 timing (+2 business days); ⩾ 12 months of forward-simulated record (backlog BL-S02)
N-19	Korea leverage / flow / semis-cycle state	Both want it; no data	Does margin debt or leveraged-ETF AUM lead KOSPI rv21 by ⩾5 days, and improve vol-target ΔMDD in ⩾3 episodes?
N-08	D&M floor 0 / 0.25 / 0.5	Split within both panels	Frozen spell definition across US and international daily WML
N-10	Reversal blocked in fast stress	Role-swapped panels	Forward-simulated reversal at 35 bp per side, with stress-day spreads from KIS quote snapshots, vs the gross edge (a read-only desk has no fill logs)
N-51	Non-inferiority tier for published effects	Professors split; executives moderate	2016–26 Japan/Europe/APxJ holdouts
N-45	Oil-shock event rule on KOSPI gross	Professors split; executives opposed	Event study on spot *and* futures, conditioned on real-yield direction
N-21	Foreign flows still lead KOSPI?	Hong & Lee vs 2026 structure	KIS investor-type flows pre/post May 2026
N-35, N-36	Regional and Korea factor priors	Professors (regional, strong); Korea priors moderate (professors) and weak (executives)	Verify on KRX stock-level data net of costs
N-39	Momentum entries ×0.5–0.7 on decision days	Professors only (FOMC days 2021–26: +0.6 vs +5.3 bp at 1.3× the risk)	QUANTGEN's own announcement-day P&L
N-59	Korean supply screens (KOSPI200/MSCI calendar, rights-offering exclusion)	Executives only	DART event study

What this means for a read-only research desk

Spec v1.0 was written as if QUANTGEN routes capital; v1.1 re-maps it onto the real application, which does not. It is an open-source, read-only research desk with "no code path that sends an order" and a simulated order book. Its rules are "decide at the close of day t, earn from day t+1" and "the gate decides, not the author" (QUANTGEN public page and architecture review, local: `00_QUANTGEN_public_page_text.txt`). Its nine-step gate is fixed before each test. It runs a

purged, embargoed 756/126-session walk-forward and requires a block-bootstrap Sharpe interval that clears zero, a deflated Sharpe of at least 0.95, and a probability-of-overfitting check. It then requires a one-sided $p \leqslant 0.10$ against the benchmark, a positive Sharpe at twice the costs, a plateau rather than a peak, and at least 60% of years positive, and it reports a 40% live haircut. Translated onto that design, the panels' integration plan fits the desk's philosophy and exposes three concrete gaps.

The translation is direct. "BUILD NOW" means the rule is computed, displayed and allowed to size strategy cards in the Strategy Lab; in spec v1.1 only a card with a PASS verdict may feed the displayed recommendation, and the other BUILD NOW cards run as descriptive books labelled with their verdicts. "SHADOW" means the rule is computed and logged as a dated, pre-registered card with a forward record, and is *withheld* as a signal, which matches the desk's existing "Withheld" labels. "DO-NOT-BUILD" maps onto REJECT cards that stay on the public record. That makes the backlog's failure archive a natural extension of "it says no, in public". The existing Macro surface computes an 18-indicator sheet with z-scores and a regime read. It is the right home for the spec's context log (R-13), on two conditions. Its z-scores must use expanding or trailing windows with publication timestamps, never present-calibrated levels. And nothing in it may feed a signal: that is the panels' shared rule that slow macro is logged context (N-30), together with the rule against present-calibrated levels (N-12, EC-E-28). The same applies to the crypto page's six-part regime score.

The first gap is **the gate itself**. The nine-step gate is built to test claims of *outperformance*: step 5 requires a one-sided $p \leqslant 0.10$ against buy-and-hold on the same days. The panels showed that a de-risk overlay's Sharpe gain cannot clear such a test in any sample QUANTGEN has. The desk already has the evidence. Its Strategy Lab card "E1 · trend switch" (unrelated to the E1 persona), a long-or-cash rule on US ETF sleeves, was rejected with an out-of-sample Sharpe of 0.84 against 0.95 and $p = 0.75$, even though it cut the drawdown from −27% to −15%: "It protects; it does not outperform, and the claim was outperformance". Every overlay both panels endorse would meet the same fate, including the Korea vol target, which turned −38.6% into −12.2%.

The consistent fix is a second, pre-registered claim type for risk control. It would be judged drawdown-first on ΔMDD, CVaR, time under water, worst-day coverage and re-entry lag. It would require Sharpe non-inferiority rather than superiority, and pass leave-one-crisis-out with no episode supplying more than half the benefit. It would also report the bull tax every month. This keeps "the gate decides", because the criteria are fixed before the test. Spec v1.1 §8.7 now defines exactly this class (CARD-P, checks PC-1..PC-7), alongside Tiers A/B/C, forward-simulated books and demotion rules. The desk's own 40% post-publication haircut points the same way as the decay the panels measured for vol scaling with a 2× cap, where the Sharpe gain fell from +0.66 to +0.26, though that decay was larger. The panel verified only the working-paper version of McLean–Pontiff's figures, however, so the exact haircut remains a convention (McLean–Pontiff WP).

The second gap is **costs**. The panels and spec v1.0 assumed a 0.35% Korean round trip, with a 0.20% sell tax; spec v1.1 adopts the desk's own costs (see the Addendum). The desk pre-registers **35 bp per side for KRX stocks**, which is 70 bp a round trip and 140 bp in the 2× stress case. Every factor result in the body of this report is gross of costs; the Addendum gives the net values. The reversal family, whose stress premium is the most cost-sensitive finding, must therefore be re-tested at the desk's own cost before any Korean reversal card is believed. The same applies to the D&M gate on Korean stock momentum, which is untested on KRX data. The Korean literature finds 12-1 momentum negative in 2000–2019, with apparent profits that are a carry-over of 1–2-month reversal (KJFS; JDQS 2021).

The third gap is **data the desk already has but does not log**. The backlog's top item (BL-U01) is a Korea leverage, flow and semis-cycle state, which needs investor-type flows, margin balances and leveraged-ETF AUM. Its KRW leg needs a true-timestamp closing rate rather than the weekly H.10 file. The desk's KIS key already reads "**USD/KRW fixing, investor flows, short-sale data**". Logging these daily from now on, with publication timestamps, is the single cheapest step toward settling the most important open question. KRX margin debt and leveraged-ETF AUM still need a separate source.

Three further implications follow from the desk's current state.

Quant Lab. The GARCH(1,1)-t fan is already labelled REJECT on SPY: VaR was breached on 6.6% of days and the PIT test failed at p 0.003. KOSPI's 2026 regime, with 21-day realised volatility above 30% on 88% of days and single-day moves of +17.91% and −12.06%, sits inside today's 504-close window but will roll out of it. R-16 therefore matters. QUANTGEN should seed a named scenario library: the 2026 Hormuz path on Dated Brent *and* futures, the July 2026 KOSPI unwind, the September 2026 rates shock, the August 2024 carry unwind and April 2025 "sell America". Fed SEP error bands (±1.4 pp on the one-year policy rate in the December 2023 SEP) can serve as uncertainty priors, rather than relying on a fitted fan whose left tail is known to be thin (Fed SEP).

Rates & FX page. The page already shows the Fama regression and withholds FX momentum and dollar carry as signals. That is exactly where the UIP evidence points: the Diebold–Li curve forecast beat a random walk at 0 of 5 maturities.

Crypto. The desk's two passing strategy cards, crypto trend on KRW-BTC and BTC-USD, match the panels' crypto rule of own trend plus volatility. The prohibited net-liquidity rule (−0.20 Sharpe in 2023–26) should be recorded as a REJECT card rather than built.

For the current state, the build order in spec §12 still applies (v1.1 milestones M0–M10): scaffolding, data layer, volatility and risk layer, the B5 estimator gate, momentum sizing, engine and events, context log, cards through the nine-step gate, the forward-simulated legs, and finally the Quant Lab. The live readings on 2 Oct 2026 are a useful day-one acceptance test:

Reading	Value
Rates shadow leg	ON in data dated 23 Sep; applies from the 25 Sep close (strict H.15)
KRW shadow leg	OFF
`hedge_credit_off`	TRUE
Korea vol weight	0.61
Pair cap	Active
D&M bear state	OFF (24-month market return +39.7%)
VIX entry block	OFF

Two more spec rules apply now. The policy-break protocol (R-18) should re-validate any macro-conditioned setting after the Warsh transition and the September hike, because "a new chair is not a new rule". The backlog's P1 list (v1.1) puts the gate and data plumbing first (the B5 vol gate, the protective-card harness, timestamped KRW history, frozen test fixtures and the macro-sheet mapping), then the shadow legs, the reversal stress test on KIS quote snapshots, Korean flows, the Korea leverage state and the Korea factor priors.

Limitations: what this evidence cannot tell you

The data are mostly **latest-vintage**. FRED macro series include later revisions; only the real-time Sahm series is a true vintage. Macro-gate backtests therefore carry a look-ahead bias that ALFRED vintages would reduce, and Ghysels–Horan–Moench show real-time data "substantially" weaken macro predictive power (GHM).

There are **no KRX stock-level, investor-flow, margin or minute data** locally. The Korea leverage state, foreign-flow regimes, Korea factor priors, execution windows and surprise-to-volatility rules are therefore untested. KOSPI's 2026 extremes come from Yahoo closes and should be reconciled with KRX official closes.

All factor tests in the body use **Ken French factors gross of costs**, and the desk's own costs are twice what the panels assumed. The Addendum gives the overlay-incremental net values; the factors' own constituent turnover is still uncosted.

Inference is **weak by construction**. The 63-day circular-block bootstrap is shorter than the 214–245-day Politis–White blocks suggested for squared returns (Politis–White). Longer blocks lower the D&M gate's z-statistic from 2.01 to 1.80. Every overlay ΔSR lies inside the envelope that is undetectable over 10–30 years. The 2022–26 period was dissected episode by episode, so it is a reported stress-test set, not a holdout. Many 2026 lessons rest on n = 1: the joint yen intervention, the mega-supply flag, the DRAM-guide deceleration and the margin-called AI fund.

The **panels are simulated personas**. Their disagreements are structured arguments over shared evidence, not independent expert surveys, and their convergence partly reflects that they read the same data labs.

A few chronology sources are **T3**. They are used only for event dating and are flagged below.

Spec v1.0's execution conventions — the 09:00–09:30 KST no-entry window, closing-auction routing and the BoK release windows — are moot for a read-only desk except as assumptions about which prices are tradable. Spec v1.1 drops closing-

auction routing and keeps the windows only as evaluator blocks on simulated entries.

Conclusion

The research changes the question QUANTGEN should ask of macro. The prior design asked *which macro state predicts returns*. Six decades of macro data, a century of factor data, two panels and one honest timestamp audit answered that nothing in QUANTGEN's sample does it reliably enough to detect. The productive question is *which state tells you the risk you are carrying, or the hedge you are counting on, has changed*. Framed that way, the most valuable "macro" inputs turn out not to be macroeconomic. They are a strategy's own volatility, Korea's own volatility and concentration, the sign of the stock–bond correlation, and a timestamp. The classic macro models survive as a reading list for interpreting central banks, not as code.

For a desk that publishes its refusals, this is an advantage. Most of the research output is a well-documented list of rules that fail, and those already have a home as REJECT cards. The one structural change it forces is in the gate. A desk built to test outperformance cannot evaluate risk control, so QUANTGEN needs a second, pre-registered, drawdown-first claim type before any overlay can earn a verdict. The most important unanswered question is local: what led Korea's 2026 deleveraging. The data to answer it are partly in QUANTGEN's KIS feed today, though not yet logged.

Addendum — implications at QUANTGEN's own costs (spec v1.1)

This report was written alongside spec v1.1, so its body uses gross numbers. Spec v1.1 re-ran the golden values at QUANTGEN's pre-registered per-side costs (KRX ETF 3 bp, KRX stock 35 bp, US shares via KIS 50 bp, each also at 2×), with strict publication timing for H.10 and H.15, and judged the de-risk rules as **protective cards** (spec §8.7). A protective card claims drawdown protection, not outperformance. It is compared with a static book that holds the overlay's average exposure, and it must pass seven checks fixed before any run: a drawdown gain of at least 3 pp at 1× and 2× costs (PC-1); lower exposure on the worst 5% of days (PC-2, ratio ⩽ 0.90); Sharpe no more than 0.10 worse, with p ⩽ 0.10 (PC-3); a benefit in at least 3 drawdown episodes with none supplying over half (PC-4); 60% of years positive (PC-5); a plateau (PC-6); and enough onsets, episodes and history (PC-7). All figures below are from `results/R4b_golden_values_net.csv` and the spec. None is a run of QUANTGEN's own gate.

Rule	Gross	At QUANTGEN's cost (2×)	Protective or card checks	Indicative verdict
Momentum own-vol scaling (US, 50 bp on both legs)	Sharpe 0.714	0.672 (0.631)	ΔSharpe vs unscaled +0.273, one-sided p ⩽ 0.001; 70% of years positive	PASS on the French series
+ Daniel–Moskowitz gate, floor 0.5	0.798	0.745 (0.691)	Add-on +0.072, p 0.018; only 7 gate onsets since 1990	INSUFFICIENT DATA
Korea 20-day vol target (KRX ETF 3 bp / KRX stock 35 bp)	0.638	0.635 / 0.605 (0.632 / 0.573)	Max DD −36.8% / −37.2% vs −41.3% for the static book: +4.4 / +4.1 pp; tail ratio 0.82; non-inferiority p 0.24 / 0.32; July 2026 supplies 59% of the benefit	INSUFFICIENT DATA
US 15% vol target (50 bp)	0.602	0.505 (0.409)	Turnover 2.52 a year costs 1.26% a year; drawdown gain +0.2 pp (−3.4 pp at 2×)	REJECT
USD/KRW leg, 0.75×	+0.032 value timing; +0.001 strict H.10	+0.029 at 3 bp; −0.008 at 35 bp	Drawdown gain +1.1 / +0.1 pp; tail ratio 0.99–1.00	REJECT (value timing); INSUFFICIENT DATA by tier under strict timing (26 onsets)
US rates leg, 0.5×	+0.038 value timing; −0.000 strict H.15	−0.103 (strict, 50 bp)	Turnover 3.60 a year costs 1.80% a year; drawdown 3.5 pp worse than the static book	REJECT

What changes in this report's conclusions.

- **The crash protection survives costs; the Sharpe case never existed.** Net of 35 bp the Korea vol target still cut the 22 Jun–30 Jul 2026 crash to −12.3% (−38.6% buy-and-hold), and paid for it with +50.5% instead of +101.1% in H1 2026. Eleven years of KOSPI cannot confirm Sharpe non-inferiority, so the verdict waits for 12 months of forward-simulated record. The KRX ETF line (3 bp) is the cheaper instrument book; it must be declared on the card before the run.
- **The two "surviving" macro legs do not survive QUANTGEN's rules.** Under strict publication timing neither has a gross edge, and neither protects: their exposure on the worst days is the same as on average days. They keep running only as labelled forward-simulated refusals, as card E1 does. The report's "+0.010 under strict H.15" was the one-business-day lag; the strict lag for a close decision is two business days.
- **Momentum own-vol scaling is the only overlay with a clear net benefit,** and the only one likely to enter the displayed recommendation once its card runs on QUANTGEN's own sleeve.
- **Daily rebalancing at 50 bp eats the US vol target.** A slower or banded estimator is a new card (backlog BL-G03), not a fix to this one.
- **The volatility estimator stays SMA20.** In the B5-style check (filtered historical simulation) it breaches a 5% VaR on 4.70% of KOSPI days, 4.99% of SPY days and 5.31% of US-market days, all within ±2 binomial SE, with PIT $p \geqslant 0.01$. EWMA(0.94) is the registered challenger. GARCH(1,1)-t on SPY fails (6.33%), as in QUANTGEN's own study.
- **Until any macro card runs,** the displayed recommendation carries only the invariants: point-in-time data, bounds, the Korea caps, the drawdown ladder and the hedge state.

Sources

Tier 1: academic, official, university course material and arXiv (T1-pre)

Source	URL
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Local textbooks (user-provided): Lee & Kim *Dynamic Macroeconomics* ch.1–9 (KU slides, v2022); Blanchard & Johnson 6e; Kirchgässner–Wolters–Hassler 2e; Dutta *Strategies and Games*	`/home/claude/quantgen_macro_data/textbooks/`
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Tier 3: flagged, used for chronology only (cited in the underlying timeline, not relied on here)

Source	Use
SHRM on the −818k benchmark; Bitcoin.com on Q2-2026 BTC; uCapital on the 17 Jul 2026 tech selloff; Baker Donelson on the Section 122 tariff rate (conflicts with Tax Foundation's 10%); Top Traders Unplugged trend-index performance	Event dating and index-performance cross-checks only; every number used above has a T1 or T2 or local source

No Wikipedia, Reddit or social-media source was used.